

Marla Eisman - Curriculum Vitae

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EDUCATION

PhD in Mechanical Engineering, Controls
University of California, Berkeley College of Engineering
GPA: 3.7/4.0

August 2025 – May 2030
Berkeley, CA

Bachelor of Science in Mechanical Engineering
University of Florida, Wertheim College of Engineering
Upper Division GPA: 3.85/4.0

August 2020 – December 2024
Gainesville, FL

PUBLICATIONS

M. Eisman, B. Lam, S. Sonnino, and F. Borrelli, “Racing a Wheeled Quadruped: Active Load Transfer Mitigation via Model Predictive Control,” *17th International Symposium on Advanced Vehicle Control (AVEC)*, 2026. Code: github.com/meisman-ucb/go2w-roll-control-mpc

Y. Liu, **M. Eisman**, L. Yang, A. Ames, and F. Borrelli, “RACER: Residual-Adaptive Closed-Loop Estimation for Sampling-Based Planning in Wheeled-Quadruped Racing,” *under review*, IEEE International Conference on Robotics and Automation (ICRA), 2027.

C.F. Nino, O. Sudhir Patil, J.C. Insinger, **M.R. Eisman**, and W.E. Dixon, “Online ResNet-Based Adaptive Control for Nonlinear Target Tracking,” *IEEE Control Systems Letters*, presented at the IEEE Conference on Decision and Control (CDC), 2025.

M.R. Eisman, “Enhanced Target Tracking Using Real-Time Residual Neural Networks,” undergraduate thesis, University of Florida, 2024.

PRESENTATIONS

Oral presentation, “Racing a Wheeled Quadruped: Active Load Transfer Mitigation via Model Predictive Control,” AVEC 2026, Tsukuba, Japan, September 2026.

RESEARCH EXPERIENCE

Model Predictive Control Laboratory (Advisor: Prof. Francesco Borrelli) Berkeley, CA
Graduate Researcher August 2025 – Present

- Developed an MPC that races a Unitree Go2W on a minimum-time raceline while using the legs to actively roll the body against lateral load transfer; validated in simulation and on hardware and released open-source (AVEC 2026).
- Co-developed a hierarchical racing stack in which a GPU-parallel MPPI planner commands an RL velocity tracker through a learned residual dynamics model, adapted sim-to-real with a low-rank update from a few hundred real samples. On hardware it completed every lap up to a 3.2m/s cap, where a kinematic-model planner completed none, and kept completing laps under sim-to-real gaps where full fine-tuning failed (RACER, under review at ICRA 2027).
- Leading a perception-driven roll control project on nonplanar terrain, building a real-time depth-camera and IMU terrain lookahead (3m ahead, 0.03° noise), a sub-millisecond convex MPC that plans body lean to minimize load transfer, and an Isaac Lab PPO locomotion policy trained on 4,096 parallel GPU environments; the full perception-MPC-RL loop runs onboard a Jetson Orin and cuts steady cornering load transfer by 77% in simulation.
- Built and maintain the lab’s ROS 2 stack for the Unitree Go2W, including a 500Hz C++ joint bridge, onboard RL policy deployment, MPC and MPPI nodes, and OptiTrack, lidar-inertial odometry and RGB-D sensing.
- Supervise undergraduate researchers on the Go2W projects, including the first author of the ICRA 2027 submission.

Nonlinear Controls and Robotics Laboratory (Advisor: Dr. Warren Dixon)

Gainesville, FL

Undergraduate Researcher

January 2023 – March 2025

- Applied nonlinear control theory and machine learning algorithms to design adaptive and robust dynamic models for multi-agent herding, tracking, and swarm dynamics with outdoor real-time learning experimentation
- Developed novel deep residual neural network architecture and Lyapunov stable controller and proved the precise correspondence between the simulation and theoretical predictions
- Created a recursive Python framework to analyze and simulate adaptive control and deep learning of multi-agent systems with seamless parameter modulation, including neural depth, learning rate, and activation functions, which produced error comparisons, function convergence graphs, and real-time dynamics visualizations
- Designed and built custom IP55-rated quadcopters and a ROS 2 framework for receiving GPS, IMU, Lidar, and other sensor data over a 5G Starlink network
- Demonstrated drone control strategies to the US Air Force for multi-agent stability in communication interruptions

Gator Motorsports (Advisor: Dr. Sean Niemi)

Gainesville, FL

Lead Engineer, Vice President

August 2020 – April 2023

- Led the research and development of UF's first electric vehicle (EV) for the Formula SAE racing competition
- Earned over \$70,000 in funding by writing grant proposals and leading 30 members to contact sponsors
- Engineered a comprehensive Matlab lap dynamics model to simulate and analyze incremental race conditions with high-order dynamics including regenerative braking
- Designed a custom 400-volt lithium-ion battery powertrain and robust liquid and air cooling system to efficiently power the vehicle and maintain safe battery temperatures
- Presented research on WUFT News (PBS) and the Engineering MAEvericks podcast discussing EVs and their role in reducing the global carbon footprint

Cognition, Action, and Perception Laboratory (Advisor: Dr. Matthew Masapollo)

Gainesville, FL

Undergraduate Researcher

August 2021 – January 2022

- Developed a Matlab-based data repository and visualization system with integrated 3D oral modeling to analyze articulatory and linguistic movements
- Developed a biomechanical soft robotic packaging solution for sensors attached to the human tongue, optimizing safety and minimizing sensor interference
- Calibrated and operated an electromagnetic articulograph during studies to collect and evaluate millisecond-by-millisecond data for cognitive and speech-based patterns

PROFESSIONAL EXPERIENCE

Vatn Systems

Bristol, RI

Autonomy Engineering Intern

March – August 2025

- Developed cost-effective autonomous underwater vehicles (AUVs) with a custom inertial navigation system for use in low-communication environments.
- Designed and implemented a model-based LQR controller coupling six-degree-of-freedom vehicle dynamics to control fins, rudder, and thruster; validated performance through Python/C simulation and real-world underwater testing.
- Collaborated with PhD researchers to achieve the company's first multi-AUV swarm demonstration, integrating acoustic communication, Doppler velocity logging, and machine learning to maintain stability under complex ocean conditions.

Space Exploration Technologies Corp. (SpaceX)

Cape Canaveral, FL

Final Integration Engineering Intern

May – August 2023

- Invented durable seat protection hardware for the Dragon capsule through finite element analysis, material testing, and convergence to critical Remove Before Flight constraints, saving \$7,900 per mission
- Formulated new hazardous propellant handling standards through advanced data analysis and automated valve leak detection, reducing engineering labor by 35 tickets per mission
- Invented tooling improvements for adhesive applications, reducing technician labor by 60 hours per campaign

Tesla, Inc.*Thermal Management Engineering Intern*

Fremont, CA

May – August 2022

- Designed novel hardware-in-loop Python program to automate and parallelize continuous critical data logging and fault detection for fixtures cycling 1000+ hours from -50 to +60 °C
- Designed, fabricated, and operated seven robust electromechanical test fixtures based on extrapolated vehicle failure reports, reducing testing timeline by nine months
- Maintained global partnerships and communication with suppliers to implement targeted reliability improvements based on failure analyses

TEACHING/VOLUNTEER EXPERIENCE

University of Florida Society of Women Engineers*Volunteer & Member*

Gainesville, FL

May 2020 – December 2024

- Educated local public elementary and middle school students at 2+ events per semester to promote STEM exposure to diverse communities
- Designed and taught accessible and interactive lessons on vibrations, the water cycle, and engineering design
- Volunteered with external events, including Alachua Habitat for Humanity to construct affordable, safe homes for community members

University of Florida Mechanical & Aerospace Engineering Department*Teaching Assistant*

Gainesville, FL

January – May 2024

- Taught engineering design, manufacturing, and analysis course as a teaching assistant for Dr. Daniel Dickrell
- Conducted reviews, held 10 office hours per week, and graded assignments and exams for 84 students

Haven Hospice*No One Dies Alone Volunteer*

Gainesville, FL

September 2023 – December 2024

- Provided 1 hour per week of care through local nonprofit for end-of-life and nursing home residents, in the No One Dies Alone program
- Played guitar and sang for individuals and groups as music therapy, read books out loud, and spent quality time with patients and families

SKILLS

Robot learning: deep RL (PPO), sim-to-real transfer, learned and residual dynamics models, low-rank adaptation, residual neural networks

Control and planning: model predictive control, MPPI and sampling-based planning, trajectory optimization, nonlinear and adaptive control, Lyapunov analysis, system identification

Perception and estimation: RGB-D terrain estimation, camera-IMU calibration, lidar-inertial odometry, motion capture, sensor fusion

Tools: Python, C++, MATLAB, PyTorch, Isaac Lab / Isaac Sim, MuJoCo, ROS 2, NVIDIA Jetson Orin, CasADi, OSQP, SolidWorks, FEA

AWARDS

- **UC Berkeley First-Year Fellowship (2025)** - Awarded one year of research funding in recognition of my outstanding graduate program application
- **Blue Origin Design Grant (2023)** - Awarded funding for Gator Motorsports team based on my proposal for design innovation and diversity/inclusion efforts
- **Formula SAE Michigan (2022)** – 3rd Overall, 1st Acceleration, 3rd Design, 3rd Autocross among 100 worldwide teams for the design, build, and manufacturing of Formula-style race car
- **Southside Businessmen’s Club Youth Achiever Scholar (2020-2024)** - Awarded \$20,000 in recognition of my community outreach and academic achievement

- **Florida Bright Futures Academic Scholar (2020-2024)** - Awarded 100% university tuition coverage in recognition of my outstanding academic performance and volunteer work